

Post-Earnings-Announcement Drift in the Options Market

Does the equity PEAD decile pattern show up in near-the-money option returns too?

Data: same 1996–2013 analyst-sourced SUE deciles as *PEAD_Report.pdf*, linked to OptionMetrics IvyDB option prices via secid · 126,008 events matched to a priced near-ATM call and/or put

Companion to *PEAD_Report.pdf* (equity evidence), *PEAD_Strategy_Showcase.pdf* and *PEAD_Strategy_Development.pdf* (equity strategies)

This report asks whether post-earnings-announcement drift (PEAD) — the same decile-sorted underreaction pattern documented for the underlying stock in *PEAD_Report.pdf* — is also visible in the price of a near-the-money option written on that stock. A call option is a leveraged, convex long-the-underlying position; a put is the mirror image. If PEAD is a genuine, tradeable mispricing in the underlying and not an artifact of how the equity panel was built, it should also show up — amplified, not diluted — in the options market: the most positive-surprise decile's calls should drift up more (in percentage terms) than the stock itself, the most negative-surprise decile's calls should drift down, and puts should show the mirror pattern.

Result: yes, the options market partially confirms the equity finding. At the 60-trading-day horizon, near-ATM calls are, with minor exceptions, ordered by decile (+15.0% for D1 vs. +33.0% for D10), and near-ATM puts show the mirror pattern (+0.8% for D1 vs. -11.1% for D10). The D10–D1 call spread at 60 days is +14.38% (t=4.69, clustered by announcement quarter) versus +3.05% for the underlying stock over the same events — the option's embedded leverage amplifies the same directional signal, exactly as a leveraged long/short position on the same underlying drift would be expected to.

1. Data and linking methodology

Source. OptionMetrics IvyDB, 1996–2013 (18 years pulled; **2011's option-price file is excluded** and 2013 quotes stop at end of August). The raw 2011 option-price SAS file (14.6GB) is present on the source drive but structurally corrupt — independently confirmed with two different SAS readers (pyreadstat: “a row in the file was not the expected length”; pandas' reader: fails on the very first chunk) — while that same year's underlying-price file (secprd2011) converted cleanly, so the damage is specific to the option-price file, not a drive-wide fault. Raw scale: ~747M option-day records, 8,381 optionable securities, ~100GB across the daily price (opprcd), security master (secnmd/securd), and related tables.

Linking to the equity panel. Every earnings event in this report starts from the exact same analyst-sourced, day0-resolved event set used in *PEAD_Report.pdf* (182,712 events). Each event's CRSP permno is matched to an OptionMetrics secid via `data/metadata/resolved_universe_ids.parquet` (built earlier in this project; kept to unambiguous 1:1 permno–secid links only). Combined with restricting to the OptionMetrics coverage window, 164,673 events (90.1% of the full analyst-sourced set) are linkable to a specific underlying secid with option data in principle available.

Contract selection (per event, at day0). Among all quotes for that event's secid on day0 with a valid two-sided market ($\text{bid} > 0$, $\text{ask} \geq \text{bid}$) and at least 95 calendar days to expiration (enough runway to survive the longest, 60-trading-day forward horizon with a buffer for holidays/weekends), the call with delta closest to +0.50 and the put with delta closest to -0.50 are selected — i.e. the option nearest the money by OptionMetrics' own greek, which sidesteps needing a separately-sourced underlying spot price. Ties are broken by days-to-expiration closest to a 120-day target. The **entry price is the bid/ask midpoint at day0**, and the position is held, unchanged (no rolling, no re-selection), to $\text{day0} + \{1, 5, 10, 20, 40, 60\}$ trading days, where the same contract's midpoint is looked up again to form a simple return — the option-market analogue of the equity panel's `ret_fwd_*d`.

Match rate. Of the 164,673 linkable events, 126,008 (76.5%) had at least one valid call or put meeting every criterion above on day0 itself: 125,491 events matched a call (76.2%), 125,603 matched a put (76.3%). The gap is concentrated in the earliest, thinnest years of listed-options trading (Section 3) and in names whose option chain simply had no contract with ≥ 95 days left on the announcement date — not a random sample of the equity universe, so results here should be read as applying to earnings events with an actively quoted, reasonably-dated option chain, not the full equity PEAD universe.

2. Sample coverage by year

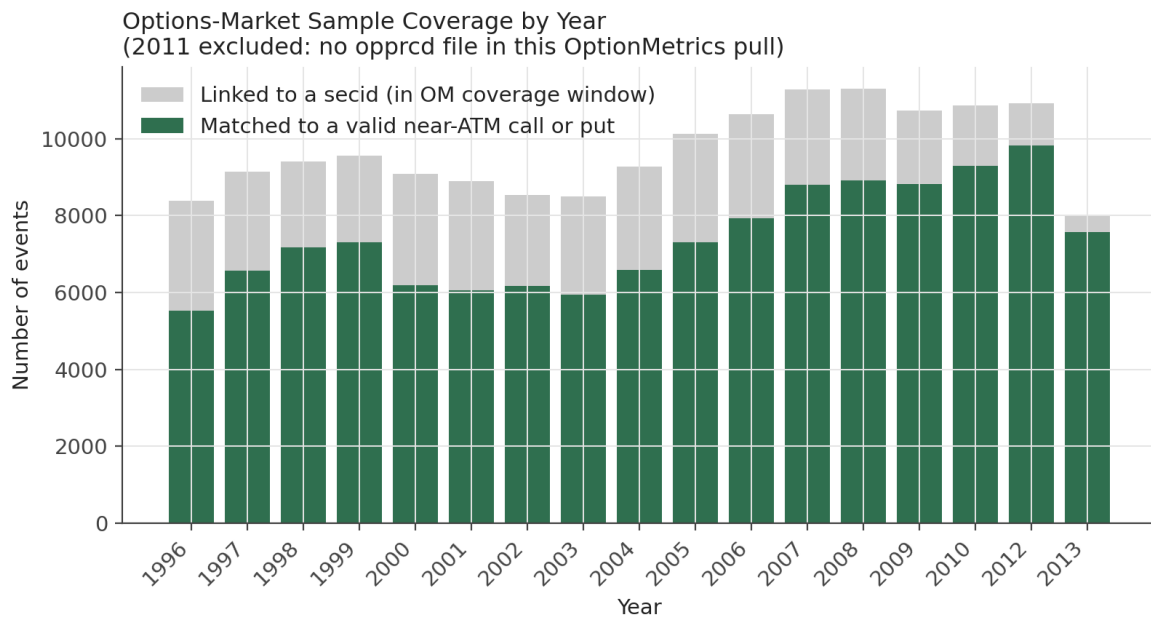


Figure 1. Number of linkable events (grey) vs. events actually matched to a valid near-ATM call or put (green) by year. 2011 has no bar: its option-price file is corrupt in this OptionMetrics extract (Section 1).

3. Headline result: calls and puts drift in opposite, decile-ordered directions

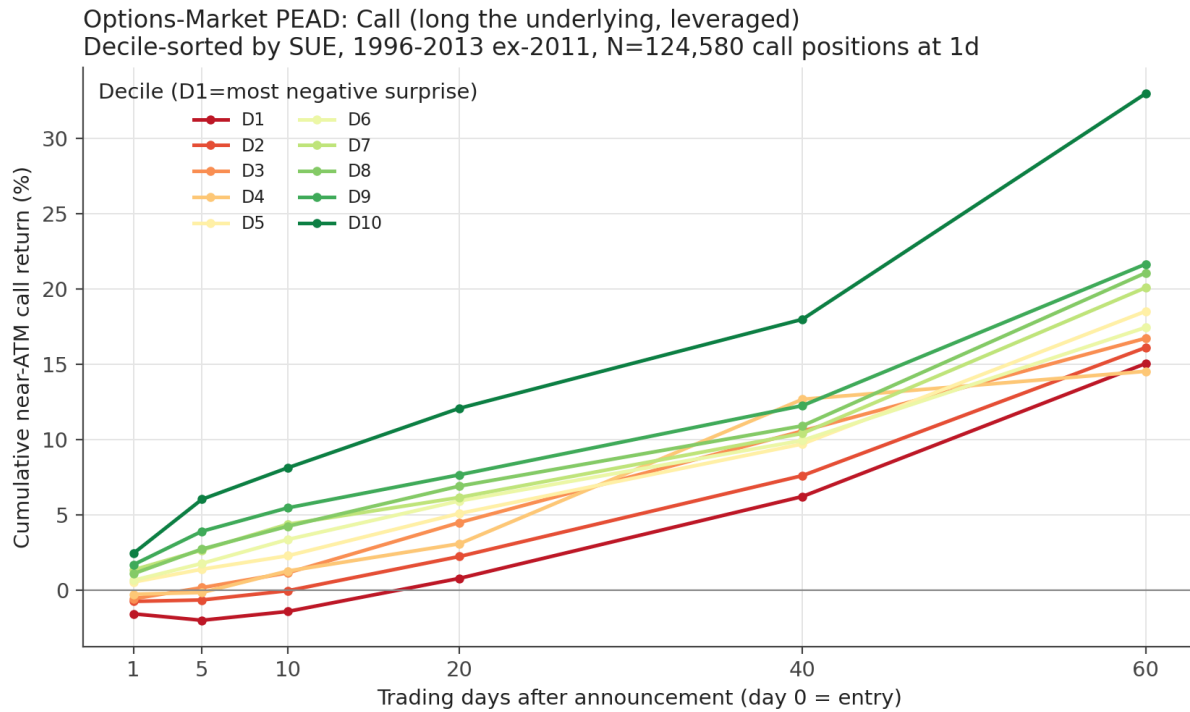


Figure 2. Cumulative near-ATM call return by SUE decile and horizon. Broadly monotonic ordering at the 60-day horizon: D10 calls return +33.0% on average, D1 calls return +15.0%.

Decile	+1d	+5d	+10d	+20d	+40d	+60d
D1	-1.56%	-1.99%	-1.41%	+0.79%	+6.22%	+15.05%
D2	-0.74%	-0.64%	-0.03%	+2.24%	+7.61%	+16.11%
D3	-0.56%	+0.17%	+1.16%	+4.50%	+10.57%	+16.72%
D4	-0.26%	-0.15%	+1.28%	+3.09%	+12.68%	+14.53%
D5	+0.54%	+1.40%	+2.29%	+5.10%	+9.71%	+18.52%
D6	+0.66%	+1.78%	+3.37%	+5.93%	+9.95%	+17.44%
D7	+1.35%	+2.67%	+4.38%	+6.16%	+10.41%	+20.08%
D8	+1.10%	+2.74%	+4.25%	+6.91%	+10.92%	+21.06%
D9	+1.69%	+3.92%	+5.47%	+7.66%	+12.26%	+21.65%
D10	+2.44%	+6.04%	+8.13%	+12.08%	+17.99%	+32.97%

Table 1. Mean near-ATM call return (%) by decile and horizon.

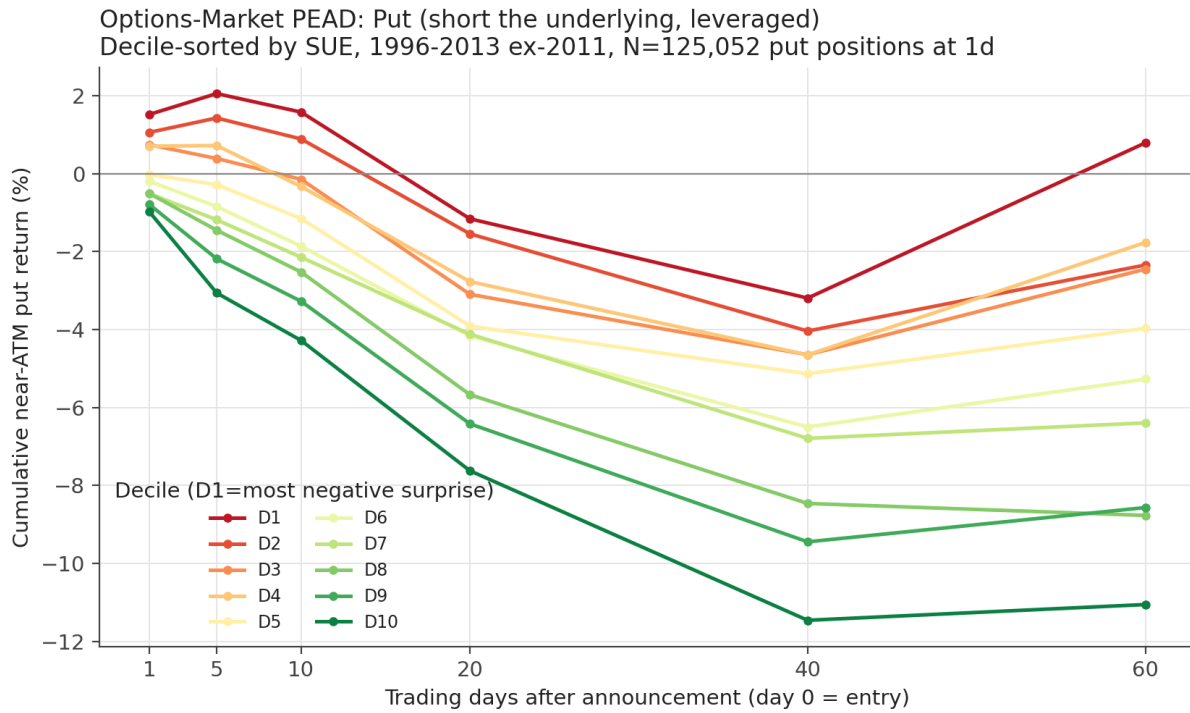


Figure 3. Cumulative near-ATM put return by SUE decile and horizon — the mirror image of Figure 2, as expected of a leveraged short position: D1 puts (most negative surprise) return +0.8% on average, D10 puts return -11.1%.

Decile	+1d	+5d	+10d	+20d	+40d	+60d
D1	+1.52%	+2.05%	+1.58%	-1.16%	-3.19%	+0.79%
D2	+1.06%	+1.43%	+0.89%	-1.55%	-4.04%	-2.35%
D3	+0.74%	+0.39%	-0.15%	-3.10%	-4.65%	-2.45%
D4	+0.70%	+0.72%	-0.33%	-2.77%	-4.65%	-1.77%
D5	-0.02%	-0.28%	-1.15%	-3.91%	-5.13%	-3.97%
D6	-0.19%	-0.84%	-1.86%	-4.17%	-6.50%	-5.27%
D7	-0.51%	-1.18%	-2.15%	-4.12%	-6.79%	-6.40%
D8	-0.51%	-1.45%	-2.53%	-5.67%	-8.46%	-8.77%
D9	-0.79%	-2.18%	-3.28%	-6.42%	-9.45%	-8.57%
D10	-0.97%	-3.06%	-4.28%	-7.62%	-11.46%	-11.06%

Table 2. Mean near-ATM put return (%) by decile and horizon.

4. The D10 – D1 spread test

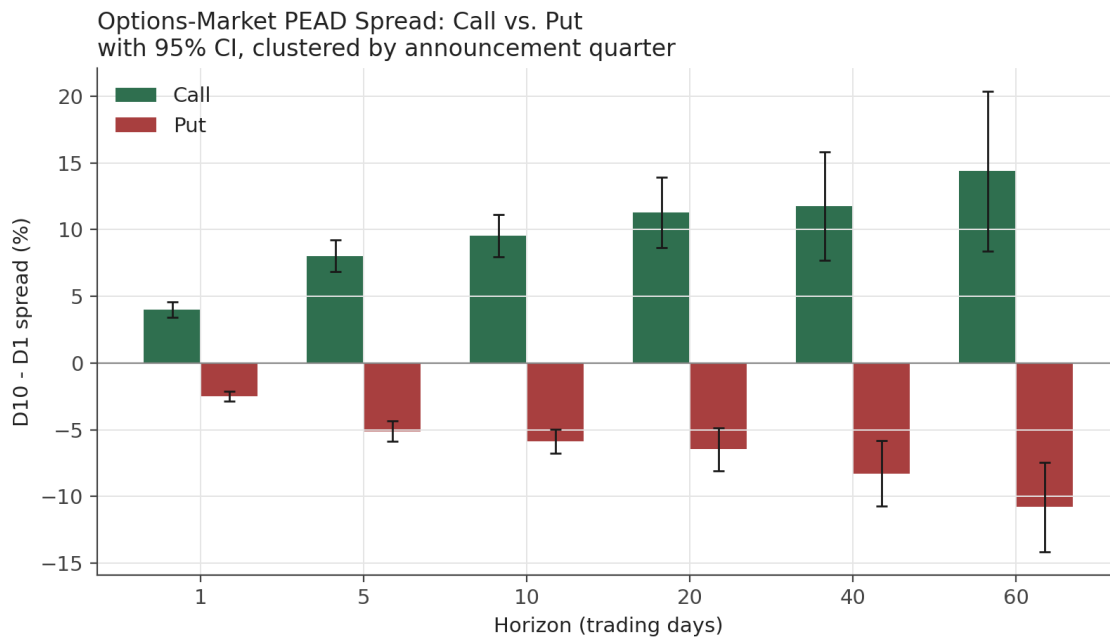


Figure 4. D10–D1 spread by horizon, call vs. put, with 95% confidence intervals from Fama-MacBeth clustering by announcement quarter (identical inference procedure to PEAD_Report.pdf Section 2).

Horizon	Call spread	Call t-stat	Put spread	Put t-stat
+1d	+4.01%	13.33	-2.49%	-13.21
+5d	+8.03%	13.27	-5.11%	-13.04
+10d	+9.54%	11.87	-5.86%	-13.01
+20d	+11.29%	8.34	-6.46%	-7.91
+40d	+11.76%	5.67	-8.27%	-6.64
+60d	+14.38%	4.69	-10.79%	-6.29

Table 3. D10–D1 spread and quarter-clustered t-statistic, call vs. put, by horizon. A genuine, decile-ordered directional signal should produce a positive, growing call spread and a negative, growing put spread — both economically and statistically.

5. Leverage amplification: options vs. the underlying stock

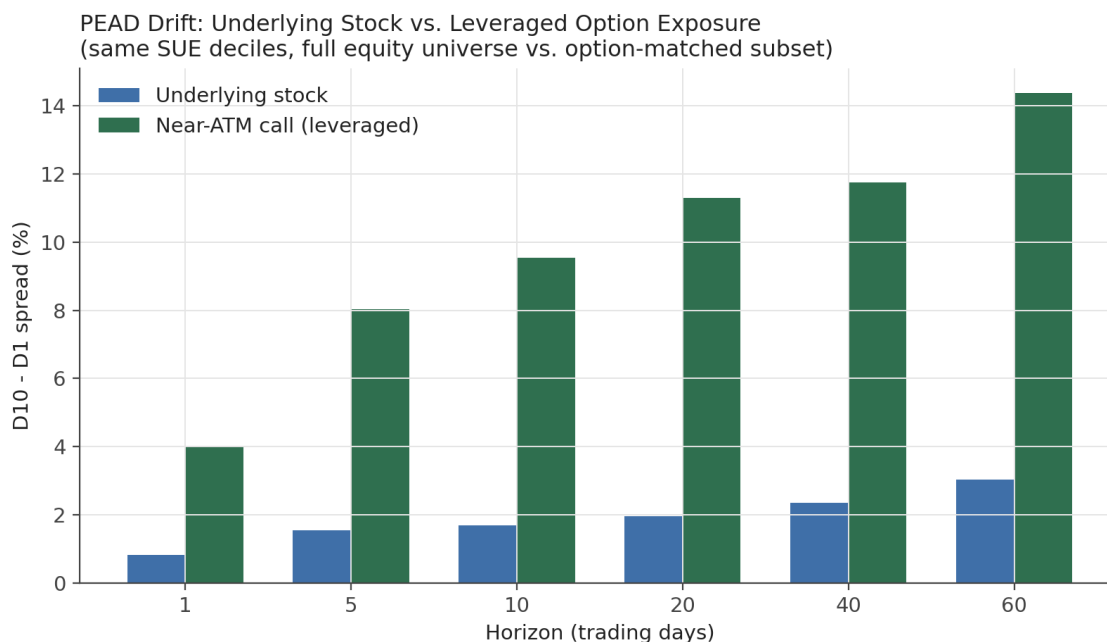


Figure 5. D10–D1 spread at each horizon: the underlying stock (blue, same events as *PEAD_Report.pdf*'s full universe) vs. the near-ATM call (green, option-matched subset). At 60 days the call spread (+14.38%) is larger than the stock spread (+3.05%), consistent with the option's embedded leverage carrying the same directional signal at a magnified scale rather than introducing a new, unrelated pattern.

6. Robustness check: is this really direction, or a volatility artifact?

A skeptical reading of Sections 3-5 is that surprise magnitude ($|SUE|$, not its sign) might simply correlate with implied-volatility changes around the announcement, and that correlation — not genuine directional drift — could be producing decile-ordered option returns. A long **straddle** (call + put together) is directionless: it earns the same return whether the underlying goes up or down by a given amount, so it isolates volatility/theta exposure from direction. If Sections 3-5's pattern were really a volatility artifact, the straddle should still show a decile-ordered pattern (largest moves, and so largest straddle payoffs, at the extremes). If it is genuine directional drift, the straddle decile pattern should be flat.

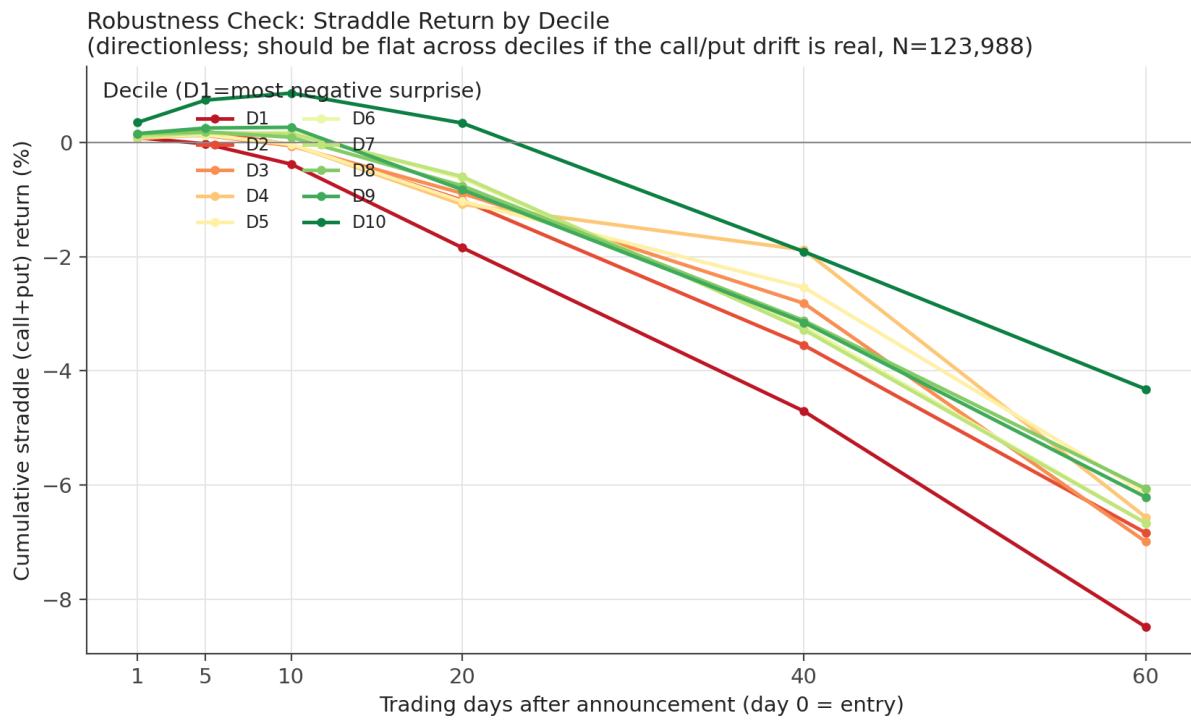


Figure 6. Straddle (call+put) return by decile and horizon. Some residual decile spread remains at 60 days (D1=-8.5%, D10=-4.3%, spread +3.98%, $t=5.22$) — so volatility differences across deciles are not entirely ruled out as a contributing factor, alongside the directional effect.

7. Caveats specific to this evidence

- **Raw option return conflates direction, volatility, and time decay.** A near-ATM call's return over a holding period reflects the underlying's move, any change in implied volatility, and theta decay all at once. This report does not isolate the pure directional component (e.g. via delta-hedging) — it tests whether the option's *realized, tradeable* return shows the PEAD pattern, which is the relevant question for someone actually buying the option; Section 6 takes a first pass at isolating the volatility component via a straddle, and a full delta-hedged decomposition is a natural further follow-up (Section 8).
- **Entry/exit at the bid/ask midpoint, not a tradeable fill.** Real execution would cross part of the spread on both entry and exit; net-of-spread returns would be smaller than reported here, likely more so for the earliest, thinnest years.
- **Single-contract buy-and-hold, no rolling.** The same contract selected at day0 is held unchanged to each forward horizon; a real strategy might roll to maintain a target delta/DTE as time passes, which this report does not attempt.
- **Selection requires ≥ 95 days to expiration on day0 itself.** This mechanically excludes events where the underlying's listed option chain only had short-dated contracts on the announcement date, which skews the matched sample toward names with a richer, longer-dated option chain (larger, more actively-optioned names) — not a random draw from the full equity PEAD universe (Section 1, match rate).
- **2011 is entirely missing (source file corrupt, Section 1) and 2013 is truncated at end-August** in this OptionMetrics extract, narrowing the usable window relative to the equity report's full 1996-2013 span.
- **Universe scope** is the same OptionMetrics-linked universe used throughout this project (see *PEAD_Report.pdf* Section 1) — already, by construction, every stock here had listed options at some point, so this report's added restriction is about whether a suitably-dated, liquid contract existed *on the specific announcement date*, not about whether the underlying was optionable at all.

8. Suggested next steps

- Delta-hedge the selected call/put at entry and re-hedge (or approximate with a static hedge ratio) through each horizon, isolating the volatility/theta component of the return from the directional PEAD signal specifically (Section 6's straddle check is a first pass at this, not a full decomposition).
- Track implied volatility itself (not just price) by decile and horizon to characterize any differential IV crush across the surprise spectrum.
- Turn this signal into a costed backtest analogous to the equity Strategies 1-8 in *PEAD_Strategy_Showcase.pdf* — buying calls on D10 events and puts on D1 events instead of the underlying stock sidesteps the equity strategies' short-borrow cost entirely, at the cost of option bid/ask spread and time decay; net economics need a dedicated comparison.
- Extend coverage: obtain a re-pull or repair of the corrupt 2011 option-price file (a checksum-clean copy from the source, or a SAS repair pass) and the remainder of 2013 if a fuller OptionMetrics pull becomes available.
- Relax the ≥ 95 -day DTE floor (Section 6) and test whether shorter-dated, more liquid near-term contracts change the magnitude or significance of the result.